

Introduction to Quantum Computing

Session 1 – Day 1

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- Comprehend the three major paradigms of quantum computing (Circuit, MBQC, Adiabatic).
- Compare different physical platforms for implementing qubits (superconducting, trapped ions, neutral atoms, photonics, and spin qubits).

Why Quantum Computing?

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Certain problems (e.g., integer factoring, quantum systems simulation) admit polynomial-time quantum algorithms, whereas unstructured search admits a quadratic quantum speedup.

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Quantum computation does not rely on a single source of speedup. It exploits:

- **Superposition:** The ability to exist in multiple states simultaneously.
- **Interference:** Amplifying the probabilities of correct answers while cancelling incorrect ones.
- **Entanglement:** Creating non-local correlations that cannot be replicated classically.

Postulates of Quantum Mechanics (Qubit-Centric)

- 1 **State Space:** The state of an isolated qubit is a unit vector in a complex Hilbert space $\mathbb{C}^2$:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1,$$

where $\{|0\rangle, |1\rangle\}$ form the orthonormal *computational basis*.

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- 3 **Measurement:** Measuring in the computational basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$. The post-measurement state collapses to $|0\rangle$ or $|1\rangle$ accordingly.

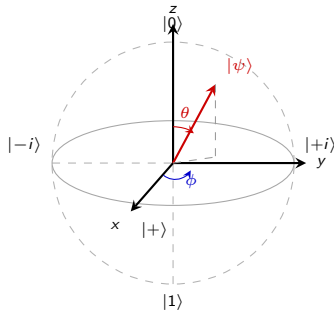
The Bloch Sphere – Visualising a Qubit

Any pure qubit state can be parameterized as:

$$|\psi\rangle = \cos\frac{\theta}{2} |0\rangle + e^{i\phi} \sin\frac{\theta}{2} |1\rangle$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

- **North Pole:** $|0\rangle$ ($\theta = 0$)
- **South Pole:** $|1\rangle$ ($\theta = \pi$)
- **Equator:** Equal superpositions (e.g. $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ at $\theta = \pi/2, \phi = 0$)
- **Phase ϕ :** Longitudinal position; crucial for interference.



Toy Example: Hadamard + Measurement

Analytic Cycle

- 1 **Preparation:** Start in the state $|\psi_0\rangle = |0\rangle$.
- 2 **Evolution:** Apply Hadamard $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$:

$$|\psi_1\rangle = H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \equiv |+\rangle.$$

Here, the amplitudes are $\alpha = 1/\sqrt{2}$ and $\beta = 1/\sqrt{2}$.

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Interpretation

The Hadamard gate maps a basis state to a coherent superposition. Measurement destroys this coherence, yielding a completely random

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- Entanglement is a vital resource for quantum communication (e.g., teleportation) and computational advantage.

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- PSPACE (Polynomial Space)**: Solvable using polynomial memory space (e.g., QBF, Generalized Geography).

Complexity Relationships

Containment chains:

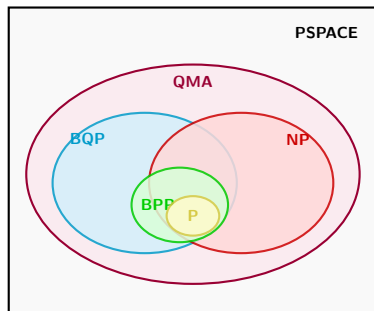
$$P \subseteq BPP \subseteq BQP \subseteq PSPACE$$

$$NP \subseteq QMA \subseteq PSPACE$$

$$BQP \subseteq QMA$$

Key Insights:

- **BPP $\subseteq$ BQP:** Quantum computers can easily simulate classical randomness.
- **NP vs BQP:** Quantum computers are not magic search engines; it is widely believed that $NP \not\subseteq BQP$



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3. **Adiabatic Quantum Computing (AQC):** Deforms an easily prepared initial ground state of H_{init} slowly to H_{problem} according to:

$$H(t) = \left(1 - \frac{t}{T}\right) H_{\text{init}} + \frac{t}{T} H_{\text{problem}}$$

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Equivalence: All three paradigms are computationally equivalent and can simulate each other with polynomial overhead.

Physical Realisation: DiVincenzo Criteria

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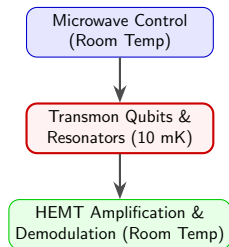
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- 5 A qubit-specific measurement capability.

Physical Realisation: Superconducting Qubits

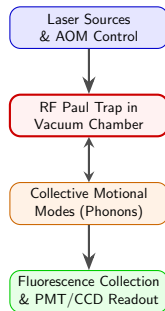
1. Superconducting Qubits

- **System:** On-chip circuits with Josephson junctions (e.g., transmons) acting as artificial atoms.
- **States:** Lowest two energy levels of anharmonic oscillator.
- **Pros:** Fast gates ($\sim \text{ns}$), fabricated using silicon lithography.
- **Cons:** Short coherence ($\sim 100 \mu\text{s}$), needs dilution fridge ($\sim 10 \text{ mK}$).
- **Players:** IBM, Google, Rigetti, IQM.



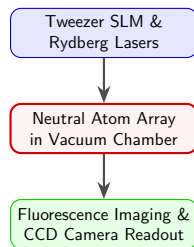
2. Trapped Ion Qubits

- **System:** Charged atoms (e.g., $^{171}\text{Yb}^+$) suspended in vacuum via RF traps (Paul traps).
- **States:** Hyperfine or stable optical electronic levels.
- **Pros:** Identical qubits, long coherence (seconds to hours), high gate fidelity.
- **Cons:** Slow gates (μs to ms), complex laser control, scaling limits.
- **Players:** Quantinuum, IonQ, Alpine.



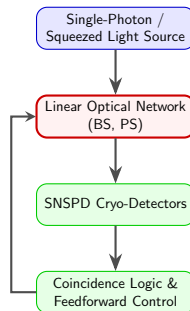
3. Neutral Atom Qubits

- **System:** Neutral atoms (e.g., ^{87}Rb) trapped in optical tweezer arrays.
- **States:** Hyperfine levels; two-qubit gates via Rydberg blockades.
- **Pros:** Scaling (thousands of qubits), reconfigurable 2D/3D grids.
- **Cons:** Slower gates, atom loss, laser power/stability requirements.
- **Players:** QuEra, Pasqal, Atom Computing.



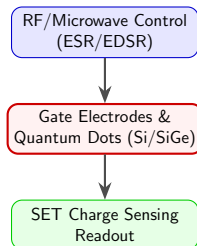
4. Photonic Qubits

- **System:** Single photons in waveguides or free space.
- **States:** Polarization, path, or time-bin.
- **Pros:** Negligible decoherence at room temp, ideal for quantum networks.
- **Cons:** No natural photon-photon interaction (two-qubit gates are probabilistic).
- **Players:** PsiQuantum, Xanadu, Quandela.



5. Semiconductor Spin Qubits

- **System:** Individual electron/hole spins in electrostatic quantum dots.
- **States:** Spin-up and spin-down in a magnetic field.
- **Pros:** Small size (sub-micron footprint), CMOS manufacturing compatibility.
- **Cons:** Charge/magnetic noise, high fabrication tolerances, sub-Kelvin (100 mK).
- **Players:** Intel, Diraq, Quantum Motion.



- 1 **Interference and Qubit Phases:** Show that relative phase ϕ is invisible in a direct Z -basis measurement of $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$, but becomes measurable as $P(0) = \frac{1}{2}(1 + \sin \theta \cos \phi)$ if we first apply a Hadamard gate H . Why is this critical?
- 2 **Why Grover's Search Does Not Put NP in BQP:** Explain why Grover's $O(\sqrt{2^n})$ query complexity search does not place NP-complete problems in BQP, even though many NP problems are search-based.
- 3 **Two-Qubit Cluster State (MBQC):** Verify that the two-qubit cluster state $|\psi_{12}\rangle = CZ(|+\rangle \otimes |+\rangle)$ is entangled.

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- The three paradigms (Circuit, MBQC, AQC) are equivalent in power but lead to different hardware strategies.
- Physical qubit realizations (Superconducting, Trapped Ions, Atoms, Photonics, Spin) present distinct engineering trade-offs between gate speed, coherence, and scalability.